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Studierichting: Informatica
Oefeningenreeks 1c nr. 46,5

$$z^4 = -4$$

$$r = 4, \quad \alpha = \pi, \quad n = 4$$

$$\rho = \sqrt[4]{4} = \sqrt{2}$$

$$\theta_k = \frac{\pi + 2k\pi}{4}, \quad k \in \{0, 1, 2, 3\}$$

$$k = 0$$

$$\theta_0 = \frac{\pi}{4}$$

$$z_0 = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$z_0 = \sqrt{2} \left(\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right)$$

$$z_0 = 1 + i$$

$$k = 1$$

$$\theta_1 = \frac{3\pi}{4}$$

$$z_1 = \sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$z_1 = \sqrt{2} \left(-\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} \right)$$

$$z_1 = -1 + i$$

$$k = 2$$

$$\theta_2 = \frac{5\pi}{4}$$

$$z_2 = \sqrt{2} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right)$$

$$z_2 = \sqrt{2} \left(-\frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2} \right)$$

$$z_2 = -1 - i$$

$$k = 3$$

$$\theta_3 = \frac{7\pi}{4}$$

$$z_3 = \sqrt{2} \left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \right)$$

$$z_3 = \sqrt{2} \left(\frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2} \right)$$

$$z_3 = 1 - i$$

References